Ramtin Soleymani | Electrical Engineering Student

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PROFILE OVERVIEW

- Self-motivated fifth-year Electrical Engineering student at Toronto Metropolitan University with strong passion for both Digital and Analog Systems.
- Focused and able to work well under pressure in both collaborative and independent tasks.
- Designed 10+ circuits in a laboratory environment with the usage of Function Generators and measured the results with an Oscilloscope.
- Studied Verilog HDL strongly learning the fundamental basics to expert level design, creating and compiling 50+ Verilog HDL examples.
- Designed and conducted network experiments within a custom-built lab environment using Ubuntu-based virtual machine in an isolated environment.
- Compiled and programmed multiple projects using VIM on the Linux OS with programs such as **C, Python, Java, and Verilog HDL**.
- Currently designing and learning Schematics & Layout through cadence virtuoso working with 0.18 micrometer channel length transistors.

RELEVANT TECHNICAL SKILLS

- **Languages:** Verilog HDL, Python, Java, C, C ++, AutoCAD, EPRI-TLWGEN2, Excel, PLS-CADD, MATLAB
- **Technologies:** Cadence Virtuoso (Schemadics & Layout), Linux OS/UNIX, Quartus Prime, Multisim, ModelSim, Ubuntu
- **Others:** Debugging, Problem-Solving, Positive Attitude, Constructive, Scripting, Documenting, Organization, Computer Architecture

EDUCATION

Bachelor of Engineering – Electrical Engineering (Co-op) *Sep. 2021- Apr. 2026(Exp.)*

Toronto Metropolitan University, Toronto, ON

Relevant Courses: **CMOS Analog Integrated Circuits, Low Power Digital Integrated Systems, Embedded Systems Design, Electronic Circuits, Microprocessor Systems, Digital Systems, Signals and Systems, Solid State Physics, Software Systems**

WORK EXPERIENCE

University Co-op Student – Hydro One Networks Inc. | AutoCAD, Microsoft Office Applications, PLS-CADD, EPRI-TLWGEN2

483 Bay Street, Toronto, ON, Canada *Aug. 2024 – Aug. 2025*

- Conducted multiple studies on high-voltage transmission systems, covering lightning, grounding, and other critical analyses.
- Demonstrated excellent time management by consistently finishing tasks ahead of deadlines.
- Developed strong communication and presentation skills by leading a technical AutoCAD session and participating in weekly team meetings.
- Applied strong analytical and research skills to investigate challenges of solar panel farms near high-voltage transmission lines, producing a detailed 20-page report.

PROJECTS

CMOS Current Mirrors | Cadence Virtuoso (Layout and Schematics), Knowledge of Transistors *Sep. 2025*

- Conducted hand calculations for **simple, cascode, and high-swing** current mirroring NMOS for width and resistance value for a given reference current value.
- Designed and simulated transistor circuits in Cadence, creating both schematic and layout drawings, and evaluated current mirror performance metrics (maximum resistance, minimum output voltage, output resistance, etc.).

Single-Port Asynchronous Read SRAM | Verilog HDL, ModelSim, Quartus Prime *July-Sep. 2025*

- Designed and implemented a 16x8 single-port RAM module in Verilog with synchronous write and asynchronous read functionality, using a bidimensional register array for memory storage.
- Validated memory behavior by simulating clocked write cycles, immediate asynchronous reads, and analyzing simulation outputs for functional correctness.

Exploring Cortex-M3 Features for Performance Efficiency | Knowledge of Embedded Systems, C, Keil uvision5, ARM Cortex-M3 *Sep. 2025*

- Debugged and enhanced an as-built code in C, modified it to conduct 3 different methods of address changing (Bit-banding, Masking, & Function).
- Optimized and intensively tested 4+ timing optimizations for different conditional execution statements in C, conducting experiment using Keil Debug mode and the ARM Cortex-3 board as well.

Linear Voltage Controlled Function Generator | Multisim, Circuit Design, Oscilloscopes, Multimeters *Oct. 2023*

- Effectively used electronic components: capacitors, resistors, op-amps, transistors, and diodes, to engineer a voltage-controlled function generator, generating square and triangular waveforms at desired frequencies. Validated performance through oscilloscope and multimeter measurements.